

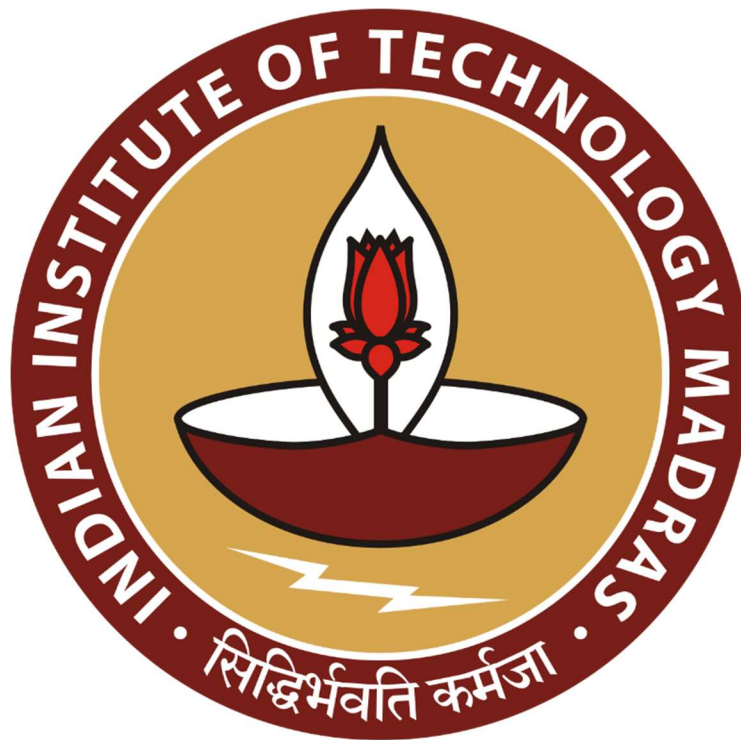
Strategic Growth and Revenue Stabilization: Enhancing Customer Retention and Inventory Management

Final report for BDM Capstone Project

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1. Executive Summary

‘S’ and ‘S’ Brothers, a key wholesale dealer in jaggery, is located in Kozhikode Big Bazaar. The business deals in B2B as well as B2C in the segment of ‘Food and Grains’. The venture has struggled with sharp revenue fluctuations due to inconsistent pricing and inefficient stock forecasting. This project applies business analytics to identify drivers of revenue instability, evaluate other problems and recommend data driven solutions for growth.

Twelve months of company sales and inventory data—including monthly product sales, revenue, and unit prices for five leading SKUs—were collected and cleaned from company ledger sheets. Analytical methods such as descriptive statistics, time series visualization, competitor benchmarking, and ABC classification were used. All transformations and calculations were performed using Excel and Python. The initial descriptive analysis quantified sales trends, revealed product-wise performance, and highlighted considerable volatility in both revenue and inventory turnover.

Key findings reveal significant revenue is concentrated in a handful of fast-moving products, while slow-movers contribute to overstocking and cash-flow constraints. Months with higher pricing fluctuations correlated with increased customer churn and reduced sales. The findings indicate that revenue shocks closely track months when S and S Brothers prices exceed those of competitors. Monthly trends, sales dips, and product priorities are outlined using visualizations. ABC analysis reveals two SKUs generate over 60% of total revenue. ABC analysis identified priority SKUs for aggressive inventory management, while time series trends informed optimal replenishment cycles and promotional timing. Based on these analysis, strategic recommendations include demand-based pricing, strict supplier vetting for quality assurance, enhanced inventory controls, and focused customer engagement strategies.

2. Detailed Explanation of Analysis Process/Method

Data Collection, Cleaning and Preprocessing

The core data for this BDM project was sourced through a series of structured and collaborative interactions with the owner of ‘**S and S Brothers**’. At the outset, a series of interviews were conducted with the owner to understand the business processes in detail — specifically focusing on sales workflow, pricing decisions, inventory management practices. With the owner’s assistance, access was gained to all available internal records. These include:

Stock Ledgers, Sales registers, Revenue and Pricing Sheets (Bill book & Day Book).

The raw data comprised transaction-level sales and inventory logs for a year. Initial steps involved consolidating daily records into a master dataset. The data sources can be accessed through the link mentioned below:

[Stock Ledger](#)

[SRU](#)

[Analysis](#)

The dataset includes daily sales and revenue figures ranging from April 2023 until March 2024.

Stock ledger consists daily sales, opening stock, closing stock, daily revenue and unit price details.

SRU consists of data concluding monthly sales, monthly revenue and average unit price of each month.

The excel sheet also contains details about competitor’s pricing information. I obtained competitor pricing information by reaching out to industry contacts through the shop owner(client). The sheet also contains details about the amount of stock returned in each month for each category. The document also contains information on ABC categorization.

Analysis consists of graphs and figures representing findings and method analogies used.

The inconsistencies in data including missing sales, zero-activity days, standardizing date formats. Multiple rounds of cross verification were performed, whereby values from different internal sources (Ledger, Day book, Sales book, Bill book) were checked for consistency and confirmed with the owner before inclusion. The top 5 SKUs studied were:

BHARATI TRADERS - Jaggery Nuts

SRI JAYA BHARATI TRADERS - Jaggery Nuts

RESHMA TRADERS - Jaggery Nuts

SRI DHARANI TRADERS - Jaggery balls

SRI VENGADESH PERUMAL TRADERS - Jaggery balls

Data Analysis Process/Method

Descriptive Statistics

Mean, median, standard deviation, minimum, maximum for monthly sales, revenue and average unit price by each SKUs were calculated.

	BHARATI TRADERS - Jaggery Nuts	SRI JAYA BHARATI TRADERS - Jaggery Nuts	RESHMA TRADERS - Jaggery Nuts	SRI DHARANI TRADERS - Jaggery balls	SRI VENGADESH PERUMAL TRADERS - Jaggery balls
count	12	12	12	12	12
mean	1339.43	1318.41	1310.28	1655.24	1405.67
std	85.00	85.60	57.52	132.92	82.48
min	0	0	1230	0	1325
25%	1245.91	1224.06	1268.31	1525	1350
50%	1300.41	1278.64	1315.59	1600	1400
75%	1390	1365.64	1333.77	1750	1400
max	1464.32	1453.24	1428.9	1850	1591

Fig 2.1: Descriptive statistics of unit price data

The above table consists of descriptive stats of monthly unit price data of each SKUs.

Sri Dharani Traders- Jaggery Balls has a highest mean unit price than other SKUs. Median (50%) is close to mean in most cases (especially for Reshma, Sri Dharani, and Sri Vengadesh), suggesting a somewhat symmetrical distribution, except for Bharati and Sri Jaya Bharati (with higher means than medians, perhaps due to a few high months). Jaggery Balls (Sri Dharani: 132.92) have high variability than Jaggery Nuts from Bharati and Sri Jaya Bharati (both ~85), but higher than Reshma (57.52), meaning Reshma's sales are most stable monthly, whereas Sri Vengadesh in Jaggery balls also has a stable monthly sale (82.48) compared to Sri Dharani. Minimum of Bharati, Sri Jaya Bharati (Jaggery Nuts), Sri Dharani (Jaggery Balls) had months with zero sales, that is why the minimum unit price for both of them were registered zero ("min" = 0). Maximum unit price and the interquartile range are higher for Jaggery Balls (Sri Dharani: 1850, Sri Vengadesh: 1591), suggesting higher peak demand possible for these SKUs.

The below Fig2.2 shows descriptive stats of monthly revenue data.

The comprehensive analysis of monthly revenue and sales data for five major SKUs reveals distinct patterns in product performance, stability, and business risk. Jaggery Balls, particularly those sold by Sri Vengadesh Perumal Traders, demonstrate consistently high average monthly revenues and a stable sales outlook, making them the strongest contributors to overall business performance. These products exhibit fewer fluctuations and rarely have months of zero sales, representing a reliable foundation for revenue planning and inventory management. Conversely, the Jaggery Nuts segment, while capable of high peak sales (as seen with Bharati

Traders), also shows significant volatility and periods of complete inactivity. High standard deviation and frequent zero-sale months suggest these products are subject to market challenges, potential competition impact. Such variability presents operational risk like overstocking, underutilized capital, and unpredictable cash flows that could undermine business stability if not properly addressed. Reshma in Jaggery Nuts segment enjoys a stable and predictable business pattern. Though Reshma's average revenue is lower than that of other SKUs in the same segment but the smaller deviation means far less volatility.

	BHARATI TRADERS - Jaggery Nuts	SRI JAYA BHARATI TRADERS - Jaggery Nuts	RESHMA TRADERS - Jaggery Nuts	SRI DHARANI TRADERS - Jaggery balls	SRI VENGADESH PERUMAL TRADERS - Jaggery balls
count	12	12	12	12	12
mean	287428.36	268255.23	186750	258853.51	458858.33
std	249354.86	193918.38	95087.57	128047.53	266659.19
min	0	0	54000	0	278600
25%	111827.5	99285	107955	154400	308850
50%	254940	250477.5	178672.5	209050	378475
75%	481658.75	440640	235781.25	403450	479175
max	807480	584340	353640	465500	1263350

Fig 2.2: Descriptive statistics of monthly revenue

	BHARATI TRADERS - Jaggery Nuts	SRI JAYA BHARATI TRADERS - Jaggery Nuts	RESHMA TRADERS - Jaggery Nuts	SRI DHARANI TRADERS - Jaggery balls	SRI VENGADESH PERUMAL TRADERS - Jaggery balls
count	12	12	12	12	12
mean	227.67	207.67	142.25	143.5	332.25
std	195.09	166.8	71.93	98.29	166.17
min	0	0	40	0	193
25%	88	76.75	82.5	96.5	224
50%	205	181.5	134	116	314
75%	332.75	321.25	188.75	254.75	360.75
max	615	463	270	274	814

Fig 2.3 Descriptive statistics of sales data

Sri Vengadesh Perumal Traders (Jaggery balls) stands out with the highest mean monthly sales (332.25) and the top maximum sales value (814), indicating strong demand and possibly broader market reach. Bharati Traders (Jaggery Nuts) and Sri Jaya Bharati (Jaggery Nuts) both maintain average sales (227.67, 207.67) but their maximums and standard deviations indicate occasional surges rather than steady high sales. Reshma Traders (Jaggery Nuts) displays the most consistent sales (lowest standard deviation, 71.93) with relatively tight interquartile ranges. This implies steady demand without sharp spikes or dips. In contrast, Bharati Traders and Sri Jaya Bharati both show high standard deviations (195.09, 166.8), revealing significant volatility in the same segment. Sri Dharani Traders (Jaggery balls) posts moderate average sales (143.5) and low variability (std:

98.29), suggesting a stable, though smaller, product line. Minimum values of zero for Bharati and Sri Jaya Bharati (both jaggery nuts) highlight operational risk some months with no sales, which could lead to stockouts or lost revenue opportunities. Quartile Analysis 75% values for Sri Vengadesh and Bharati Traders are markedly higher than their medians (50%), indicating potential for high sales months but also pointing to an uneven distribution (skewed towards higher values in some months).

Time Series Analysis

Sales, revenue and price data were visualized across time using line and bar charts to identify seasonality and volatility. This approach clarified months of peak versus low demand, supporting targeted interventions.

Time Series Analysis using both Python and Excel for each product category on a one-year period. Line plots and Bar charts were used for visualization.

The jaggery market is highly subject to seasonality (festivals, harvest cycles, weather variations). By plotting sales trends month-wise for each product/SKU, periodic demand spikes (like the July peak for Bharati Traders) and off-season slumps (e.g., February–March drops) can be uncovered. This helps in forecasting demand, ensuring timely procurement and avoiding stockouts or overstocking, both major sources of revenue instability. Sales trends allows to evaluate the real effect of strategy changes (like price adjustments) and external shocks (competitor pricing,). A visual record of sales before and after an event quantifies its business impact, so you can justify decisions and pivot quickly. Comparing sales trends across all SKUs highlights: best-performing products, volatile or declining lines. This guides to focus working capital, shelf space, and promotions on high-impact SKUs directly reducing inventory inefficiency and wasted capital.

ABC Analysis Product Classification

ABC analysis is a classic inventory categorization technique used in business management to prioritize and manage items based on their contribution to overall business value, either sales volume or revenue.

A Category: The most important segment (typically top 70–80% of value), demanding the tightest management focus.

B Category: Moderately important (next 15–20%).

C Category: Lowest contribution (bottom 5–10%), representing many items with low individual importance.

This helps the business to focus effort and resources on the “vital few” rather than “trivial many”. Optimize stock levels, reduce working capital, and streamline procurement/marketing.

The ABC analysis was calculated by:

Step 1: Calculate Total Sales/Revenue by SKU

For each SKU, total sales (or revenue) over the year were calculated. The ratio of total sales of each SKU to the total sales of whole SKU's were calculated. Similarly total revenue was also calculated.

Step 2: Rank SKUs in Descending Order

SKUs were sorted from highest to lowest by total annual sales and by total annual revenue.

Step 3: Compute Cumulative %

For each SKU, cumulative % of total sales/revenue is calculated:

Step 4: Categorize as A, B, or C

“A” category: SKUs cumulatively up to 70% of total sales or revenue.

“B” category: Next SKUs up to 90% cumulative.

“C” category: Remaining SKUs up to 10% of sales/revenue.

This business problem includes inefficient inventory and overstocking; ABC helps prioritize attention and capital. Revenue instability is often linked to stock-outs in ‘A’ item, or holding excess of C/B items that don’t move.

Competitor Pricing Analysis

For each of the main products (SKUs), competitor prices were collected monthly, via direct market intelligence by reaching out to industry contacts through the shop owner(client) or as reported by local rivals. These prices were tabulated alongside the company's average unit selling price. The ratio of difference was calculated as,
For each product category in each month:

$$((\text{Client Unit Price} - \text{Competitor Unit Price}) / \text{Competitor Unit Price}) * 100$$

Overlaying the price difference (%) line and the monthly sales volume, clear and immediate inverse relationships emerge. Even small gaps (2–3%) where our price is higher than competitors result in disproportionate declines in sales volume, confirming significant price sensitivity in this market. This has a huge impact on customer retention.

Stock Return Analysis

Stock returns indicate the quantity of jaggery products that were brought back from customers, often due to defects, unsold surplus, or quality issues.

The return rate is calculated as:

$$\text{Return Percentage} = (\text{Stock Returned} / \text{Monthly Sales}) * 100$$

Stock return analysis in your data identifies exactly where inventory and quality issues erode both trust and profitability. This helps to focus on suppliers and products with high, recurring returns and act decisively. Through supplier management and quality control, will minimize wastage, improve customer loyalty, and directly boost revenue stability.

3. Results and Findings

Time Series Sales Analysis

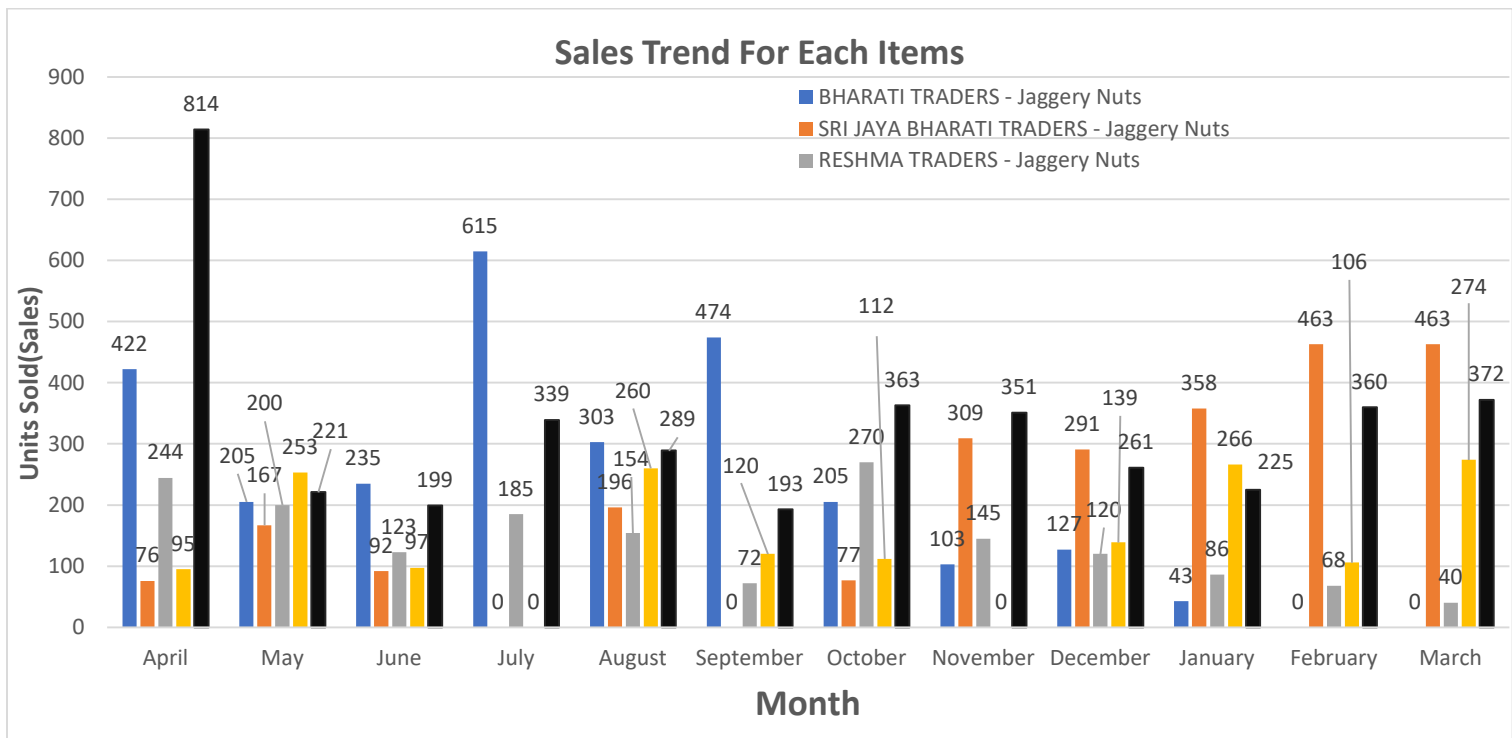


Fig 3.1 Time Series Analysis of sales for each product by each month.

Bharati Traders – Jaggery Nuts

July: Exceptionally high sales (615 kg), with elevated levels in April and September as well. Likely driven by seasonal festival demand. However, this correlation is not sustained, indicating that spikes are not due to a structured marketing strategy but more to transient market events. The absence of a continuing marketing push or customer retention effort means these peak months aren't leveraged to build lasting customer relationships, contributing to revenue instability.

October–March: Sharp, sustained decline culminating in zero sales in February and March. Despite having opening stock (confirmed from inventory data), sales failed, suggesting a probable mismatch between pricing and market (possibly out-priced by competitors in those months), or ineffective engagement after the festival

season. Reflects failure to adapt to competitor price strategies and weak market communication, exposing gaps in both price agility and customer follow-up.

SRI JAYA BHARATI Traders – Jaggery Nuts (SJB)

February–March and October–December: Sales show pronounced spikes in these late months. Could be due to less price offered than the competitors. Temple festival season in Kerala may have also contributed to the sales in October-December.

July and September: Zero sales, may coincide with weaker presence of that product category during the peak season. The competitor pricing info also reports no sales in these seasons for this product category.

Reshma Traders – Jaggery Nuts

Relatively consistent, moderate sales with minor ups and downs—contrast to the volatility observed for Bharati and SJB. Reflects a loyal, if limited, customer base but suggests weak growth. Repeated “no-sales days” even with stock (Fig 3.2) point to untapped potential, possibly due to lack of outreach or marketing. Underlines missed opportunities to expand and possible customer hesitancy due to inconsistent quality perceptions.

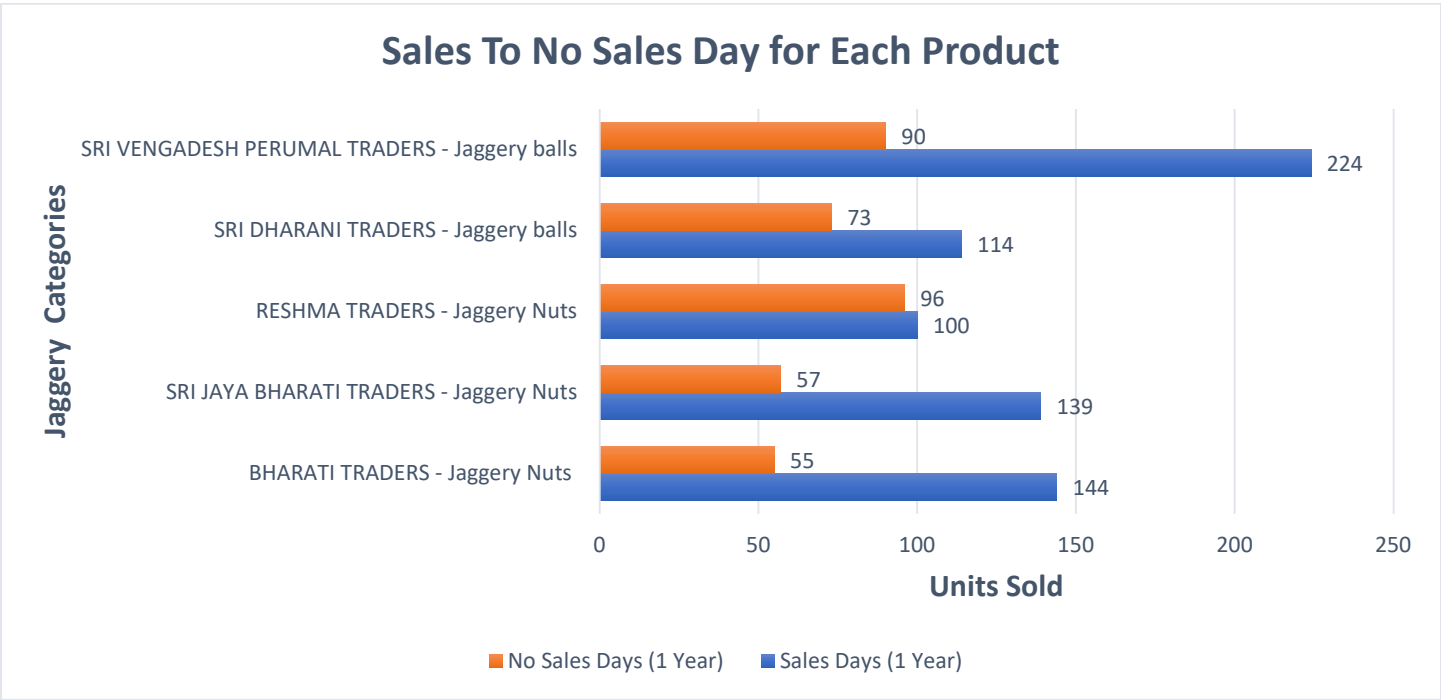


Fig 3.2

SRI DHARANI Traders – Jaggery Ball

January and May: Marked increases, aligning with periods where competitor pricing matches or is higher. Sales grow when price is competitive; dips (e.g., July, November) occur when own prices are higher. Exhibits

vulnerability to competitor price wars; also, the absence of dips in the off-season could be related to lower variation in jaggery ball quality/price compared to other brand (Sri Vengadesh Perumal) in the same category.

SRI VENGADESH PERUMAL Traders – Jaggery Balls

No zero-sales months, high median and mean sales. Suggests strong supplier reputation, consistent product quality, and possibly better customer relations or a less price-sensitive customer base. This product is least affected by the structural marketing and quality shortcomings seen in others, highlighting stable quality (not affected by artificial adulteration or supplier gaps), and possibly better customer relationship management.

Time Series Revenue Analysis

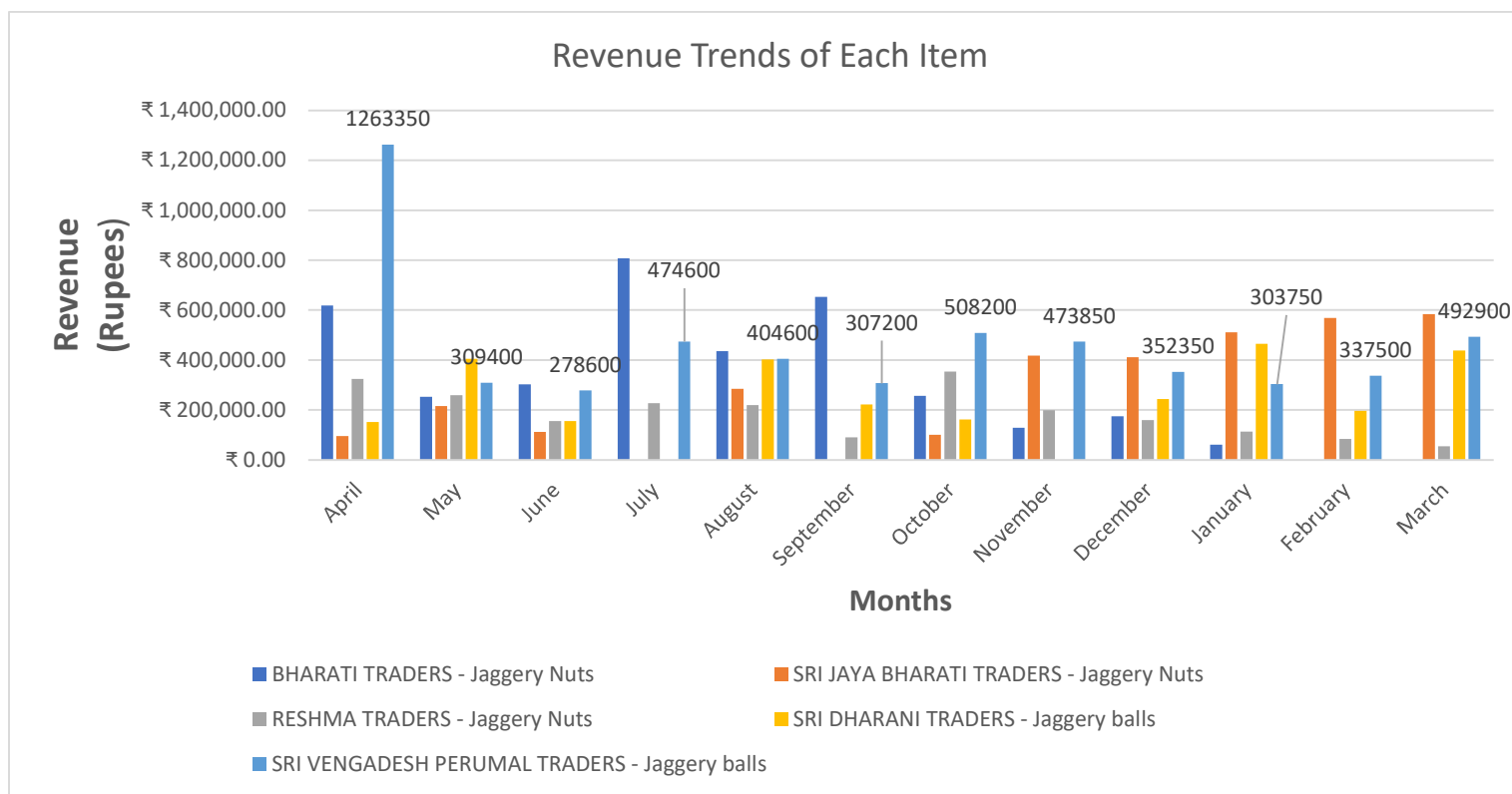


Fig3.3 Revenue Trend for each product by each month

BHARATI Traders – Jaggery Nuts

There is a strong revenue peak in July (₹807,480), corresponding to the highest sales month (615kg), and sustained high revenues in April, September, and August. However, revenue declines rapidly after September: October (₹256,980), November (₹128,750), and then drops to almost nothing by January (₹61,060), before hitting zero in February and March, which also connects with the sales analysis mentioned above. High seasonality and volatility.

Yearly Revenue: ₹3,691,675

Trend: Upward spike mid-year, then pronounced decline and business risk in the last quarter.

SRI JAYA BHARATI Traders – Jaggery Nuts

Revenue is quite erratic. There are mid-year spikes (August: ₹284,835), long zero periods (July, September), then a series of steadily increasing months from October to March (March: ₹584,340, the highest. Positively, the latter half of the year shows strong recovery and possible capture of market void left by competitors. The volatility, though, highlights risk of unpredictable cash flows.

Yearly Revenue: ₹3,300,895

Trend: Multiple months of no revenue interspersed with sharp recoveries, notably strong in last quarter.

RESHMA Traders – Jaggery Nuts

Revenue is consistently moderate with most months between ₹90,000 and ₹353,640 (October being the highest). There's a mild decline from January (₹113,640) through March (₹54,000). Reflects a stable but unspectacular performance; not exposed to wild swings or spikes, supporting stable cash flow.

Yearly Revenue: ₹2,241,000

Trend: Flat-to-slightly-declining with little monthly volatility.

SRI DHARANI Traders – Jaggery Balls

Revenue is patchy high in May (₹404,800) and January (₹465,500), completely absent in some months (July, November). December and March are also healthy (above ₹400,000). When selling, the revenue per transaction is high, but price gaps expose business to risk.

Yearly Revenue: ₹2,842,650

Trend: Intermittent peaks with repeated revenue gaps.

SRI VENGADESH PERUMAL Traders – Jaggery Balls

This is the most robust and consistent performer. Revenue rarely dips below ₹300,000/month and remains above ₹400,000 for most of the year, peaking in April (₹1,263,350) and March (₹492,900). Provides a reliable and high baseline for the business. Slight seasonality but no zero months, maintaining cash flow.

Yearly Revenue: ₹5,506,300

Trend: Sustained high revenue, no volatility.

ABC Categorization

Based on Sales and Revenue:

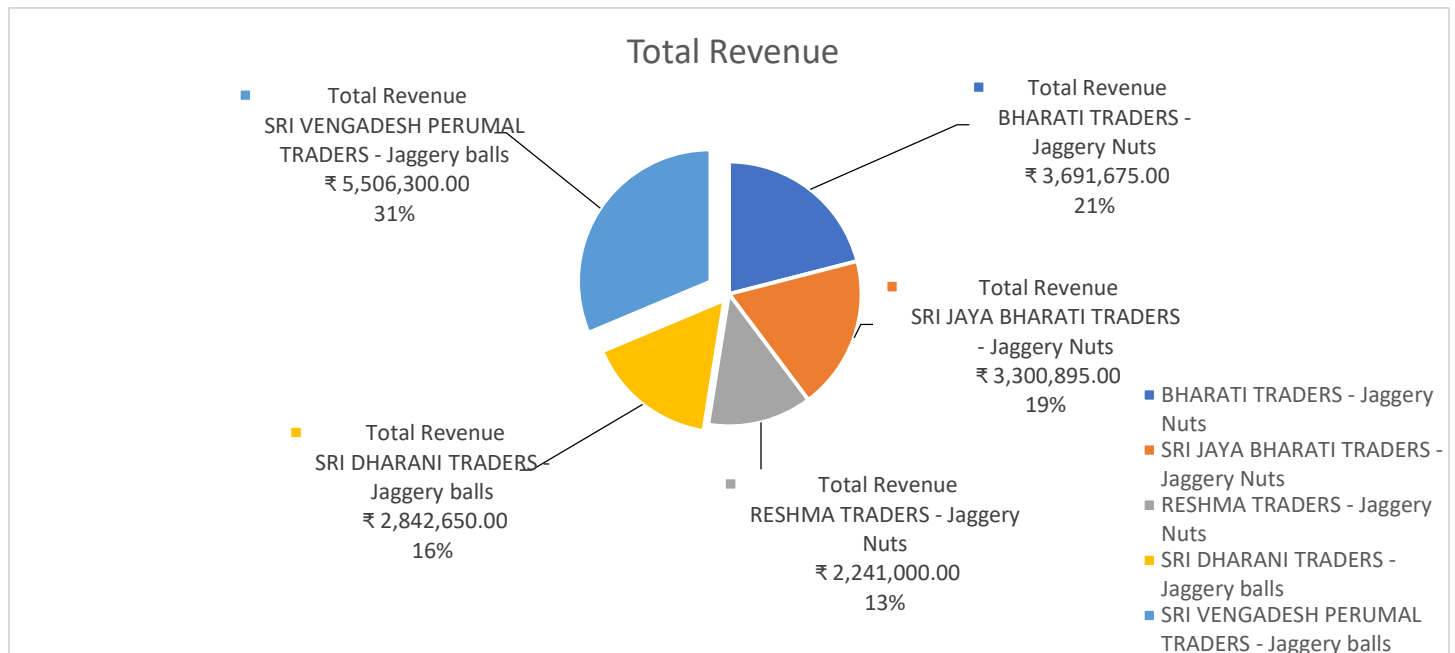


Fig 3.4

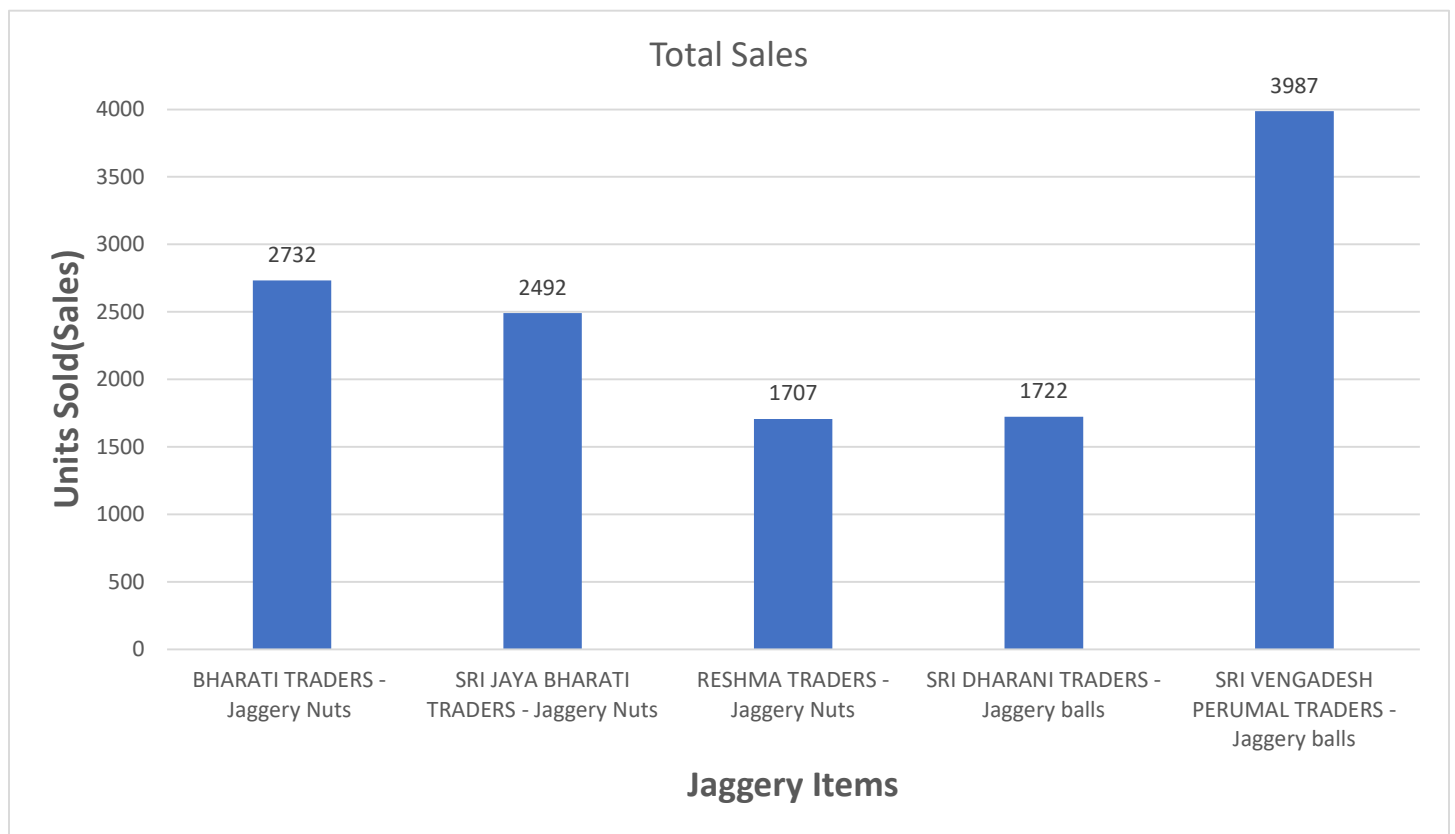


Fig 3.5

“A” category: SKUs cumulatively up to 70% of total sales or revenue.

“B” category: Next SKUs up to 90% cumulative.

“C” category: Remaining SKUs up to 10% of sales/revenue.

The overall pattern is consistent with the sales volume table, with SRI VENGADESH PERUMAL Jaggery balls being the largest single contributor by revenue (31%). Together, SRI VENGADESH and BHARATI traders constitute 52% of the total revenue and are classified as Category A.

SRI JAYA BHARATI and SRI DHARANI remain in Category B, providing an additional 35% revenue cumulatively.

RESHMA TRADERS again falls into Category C, contributing the smallest revenue share (13%)

SRI VENGADESH PERUMAL (Jaggery balls) accounts for the highest sales volume at 3,987 units (32% of total).

Along with BHARATI TRADERS (Jaggery nuts), these two constitute 53% of the total sales volume and fall into Category A, indicating high importance.

SRI JAYA BHARATI (Jaggery nuts) and SRI DHARANI (Jaggery balls) fall into Category B, together making up from 53% up to 86% of cumulative sales.

RESHMA TRADERS (Jaggery nuts) is the lowest in sales volume, contributing 14% and classified as Category C, meaning it is the least critical by sales volume.

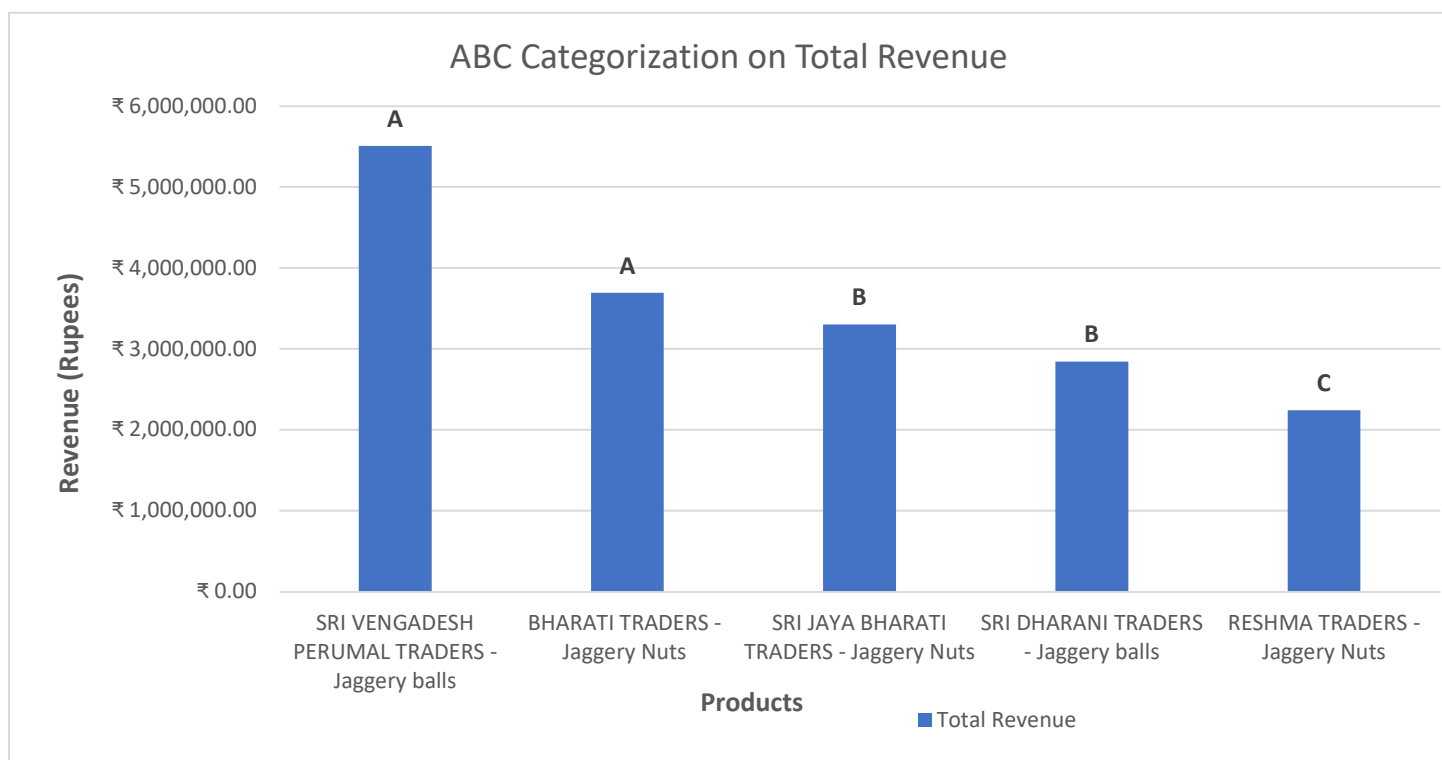


Fig 3.6

The A category (Sri Vengadesh and Bharati products) accounts for over 50% of both total sales and total revenue with just two products. These require the most stringent inventory control, price monitoring, and promotion focus.

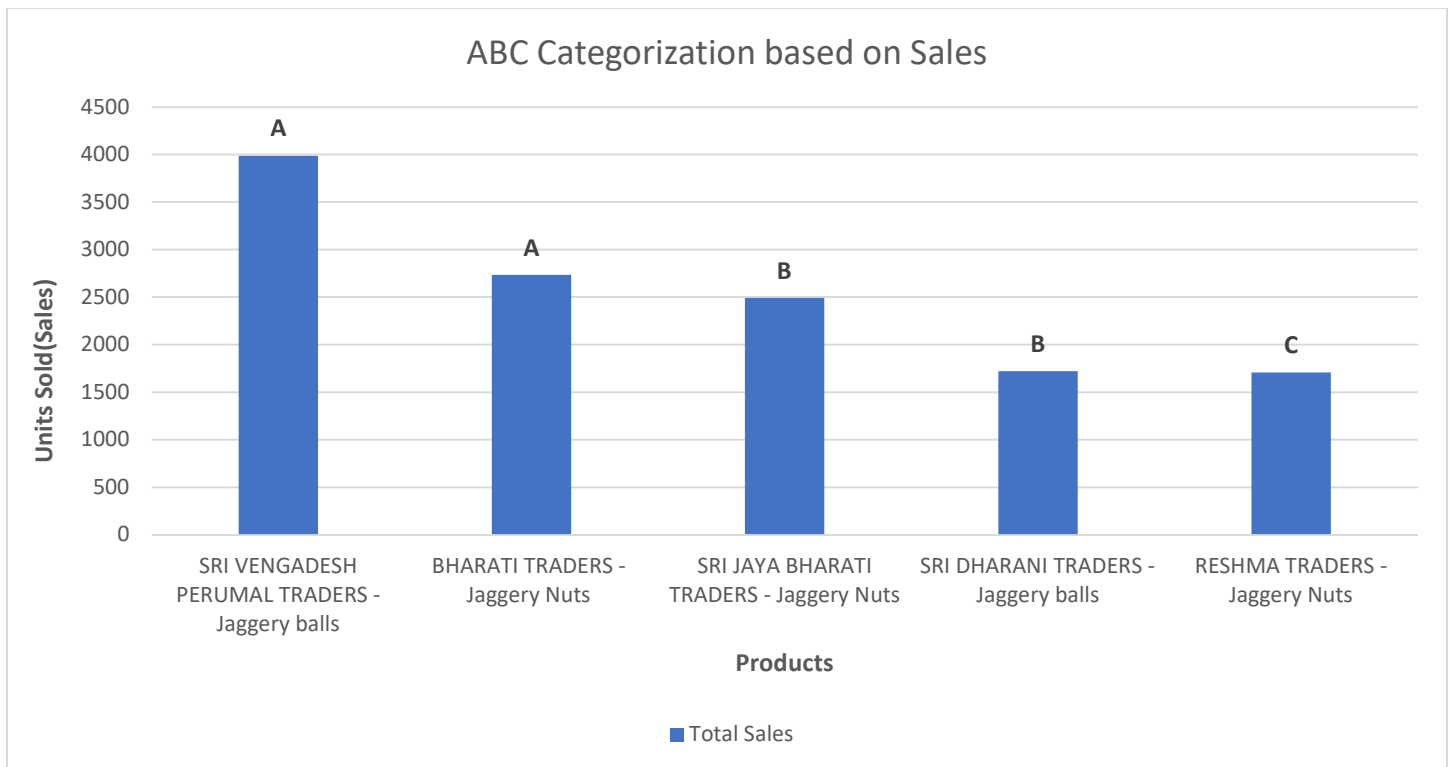


Fig 3.7

The B category (SJB and Sri Dharani) adds substantial value but with moderately less impact—these should be actively managed but may allow for a little more flexibility in stock and promotional strategies.

The C category (Reshma Traders) contributes the least—many "C" products often have slow movement, tying up capital and shelf space, and may be candidates for reduction or special clearance efforts.

Competitor Pricing Analysis

Competitor pricing analysis has been done on 3 SKU's showing high variability.

Bharati Traders – Jaggery Nuts

Throughout most months, Bharati Traders (fig 3.7) prices closely track their main competitor, often with a difference of 2% high or low.

April - August: Prices slightly lower than the competitor. This coincides with the sales data (fig 3.8) as Bharati traders registers most sales in these months.

September: Shows no competitor pricing, which shows a small spike in sales during that month.

October-January: Client Pricing higher than that of competitor's, hence decline in sales.

February and March: No client pricing, only competitor pricing appears, possibly contributing to zero sales (stock-outs or lost demand due to high market prices).

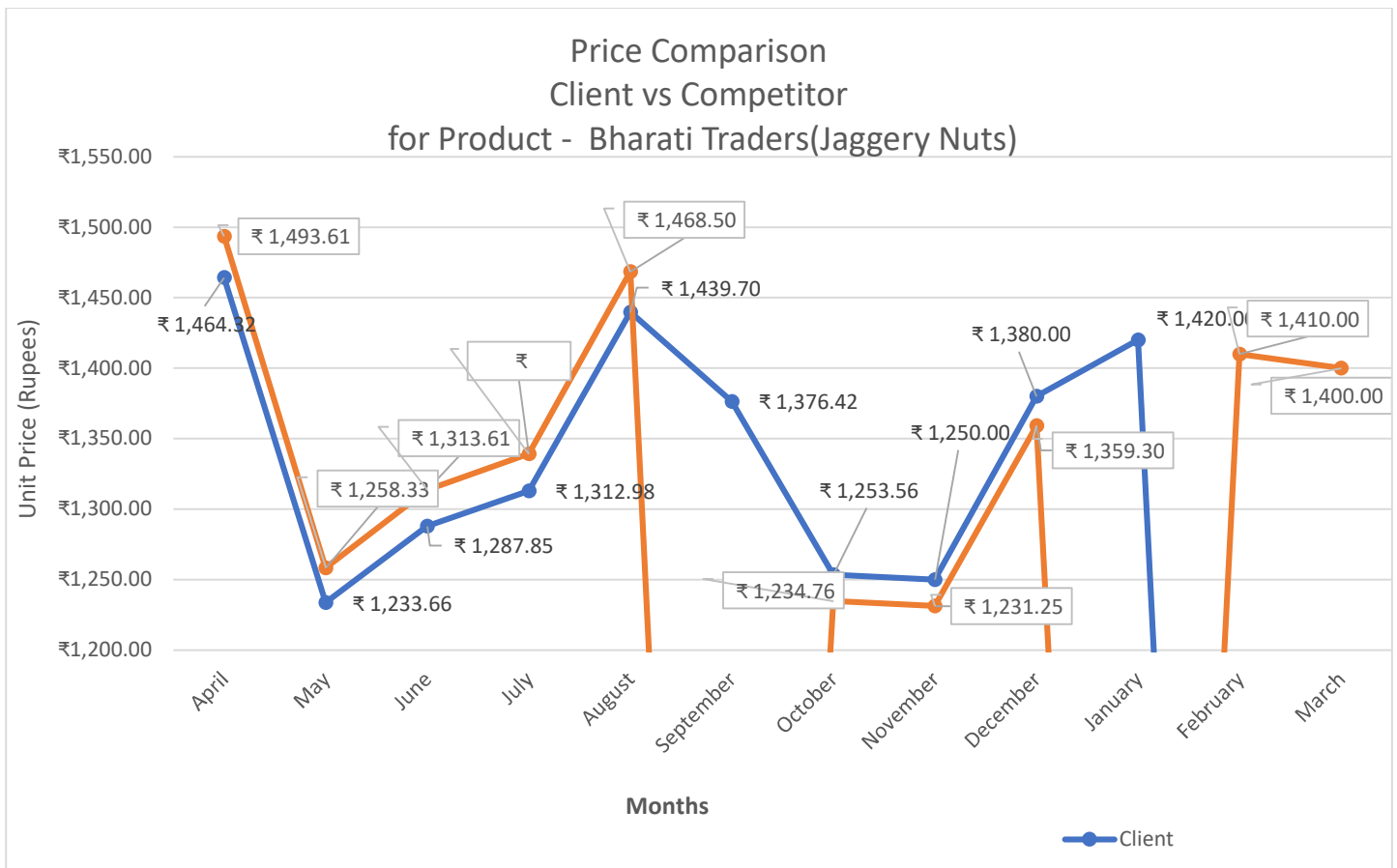


Fig 3.8

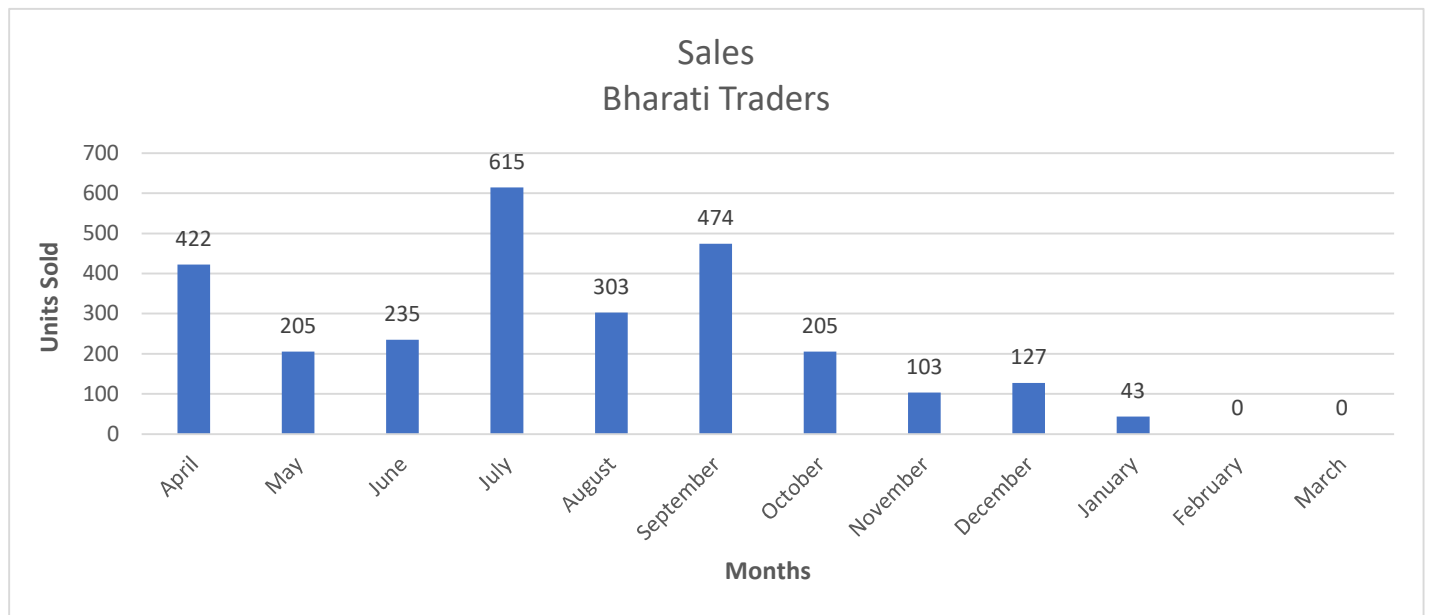


Fig 3.9

Highlights the business issue of price-sensitive customers, when pricing is not competitive, sales drop.

Sri Jaya Bharati Traders – Jaggery Nuts

Their client price trend closely mirrors the competitor price each month, sometimes marginally higher or lower

(difference 2 to 3%).

In the months April – June, client prices are higher than competitor's.

Issues with supplier side of the brand (e.g., July, September) coincides with zero sales for both client and competitor.

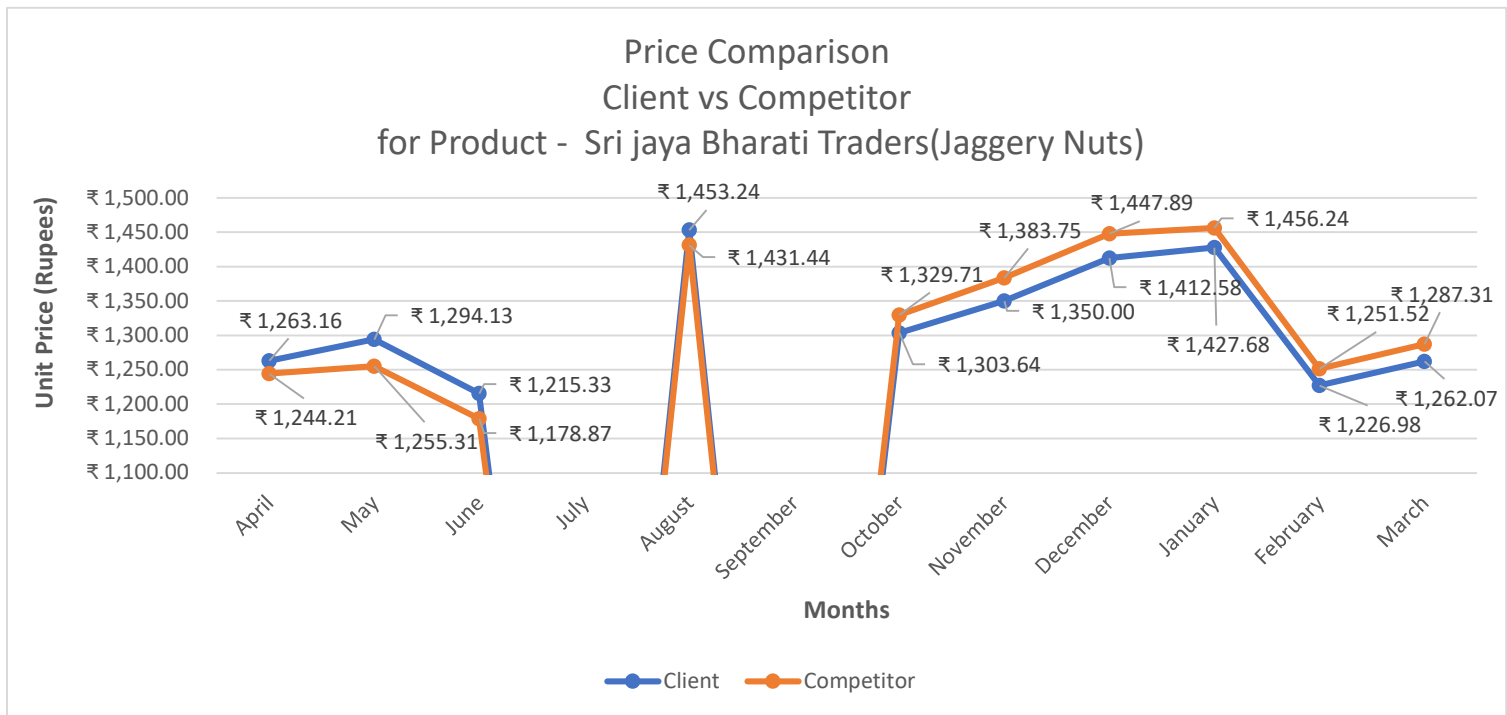


Fig 3.10

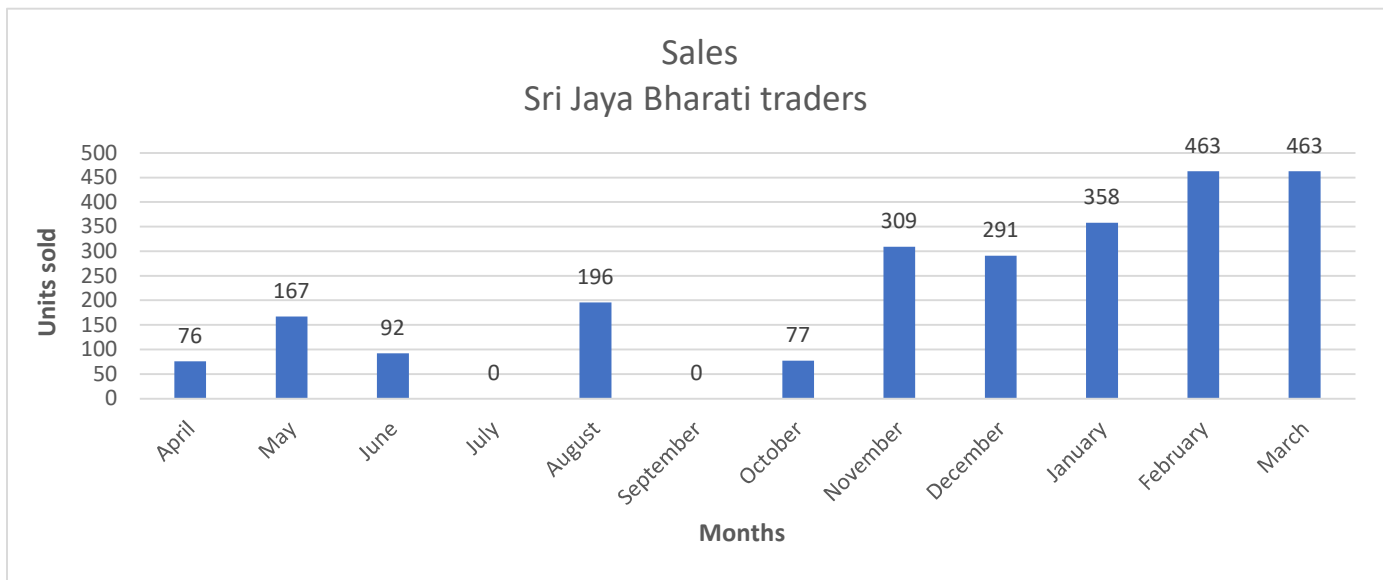


Fig 3.11

November – March: client price undercuts competitor's hence a rising trend in the sales has been observed. Demonstrates that price competition drives sales, without active price positioning, SJB is vulnerable to being overlooked even if products are available.

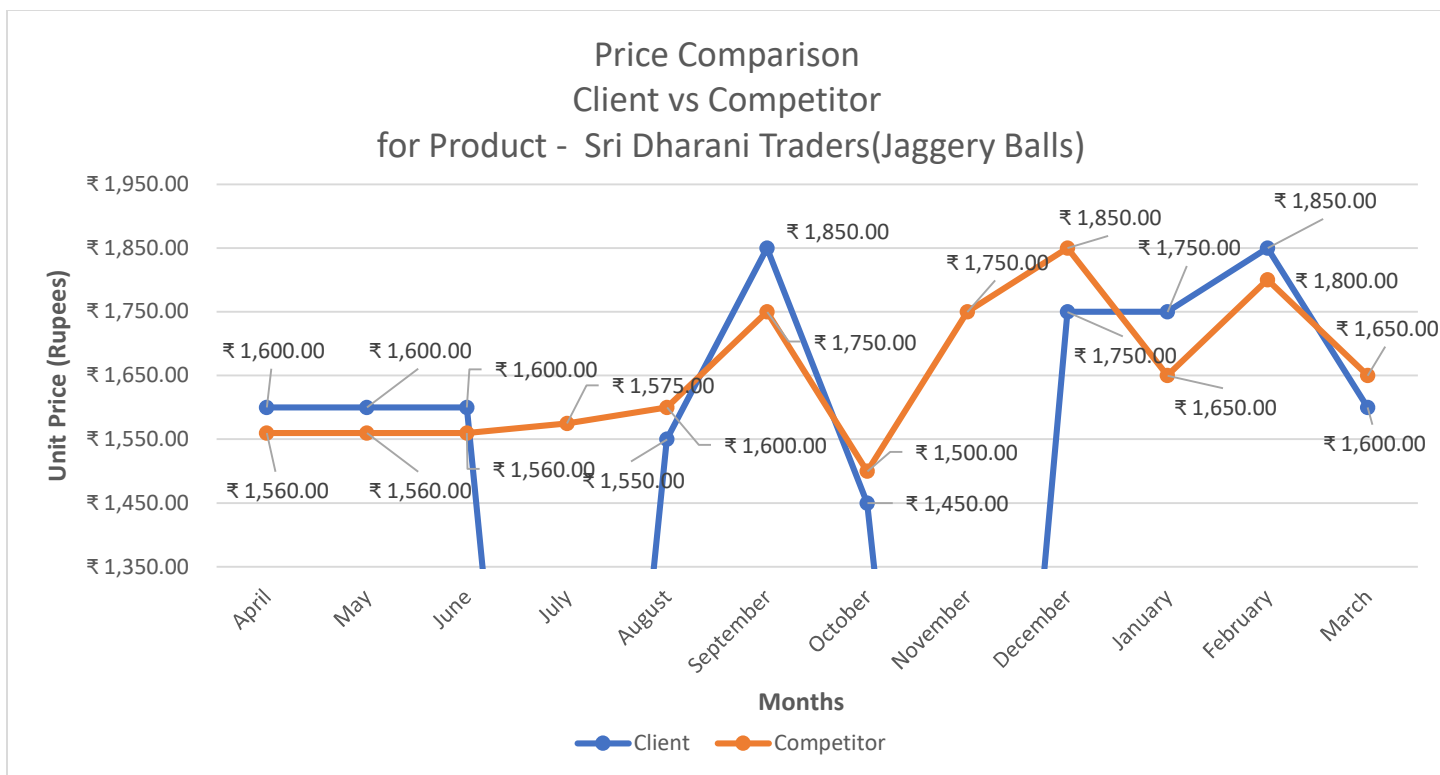


Fig 3.12

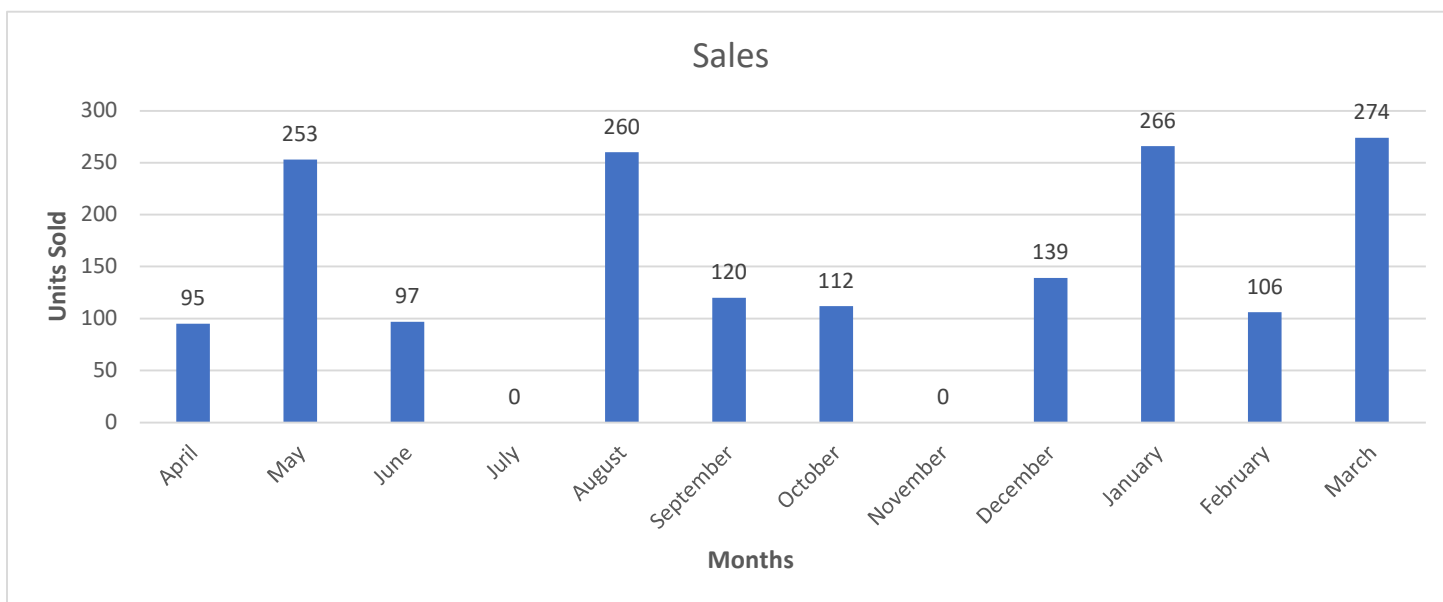


Fig 3.13

Unlike the other two SKUs this one has higher marginal differences(up to 6%).

April – June: client maintains a higher pricing than competitor, the fluctuation in sales does not coincide with pricing, suggesting seasonal sales.

The rest of the months does not show such pattern in relation to the pricing data.

This suggests demand may be more supply-driven during festival/off-season periods, but unsustainable when prices remain high.

Stock Return Analysis

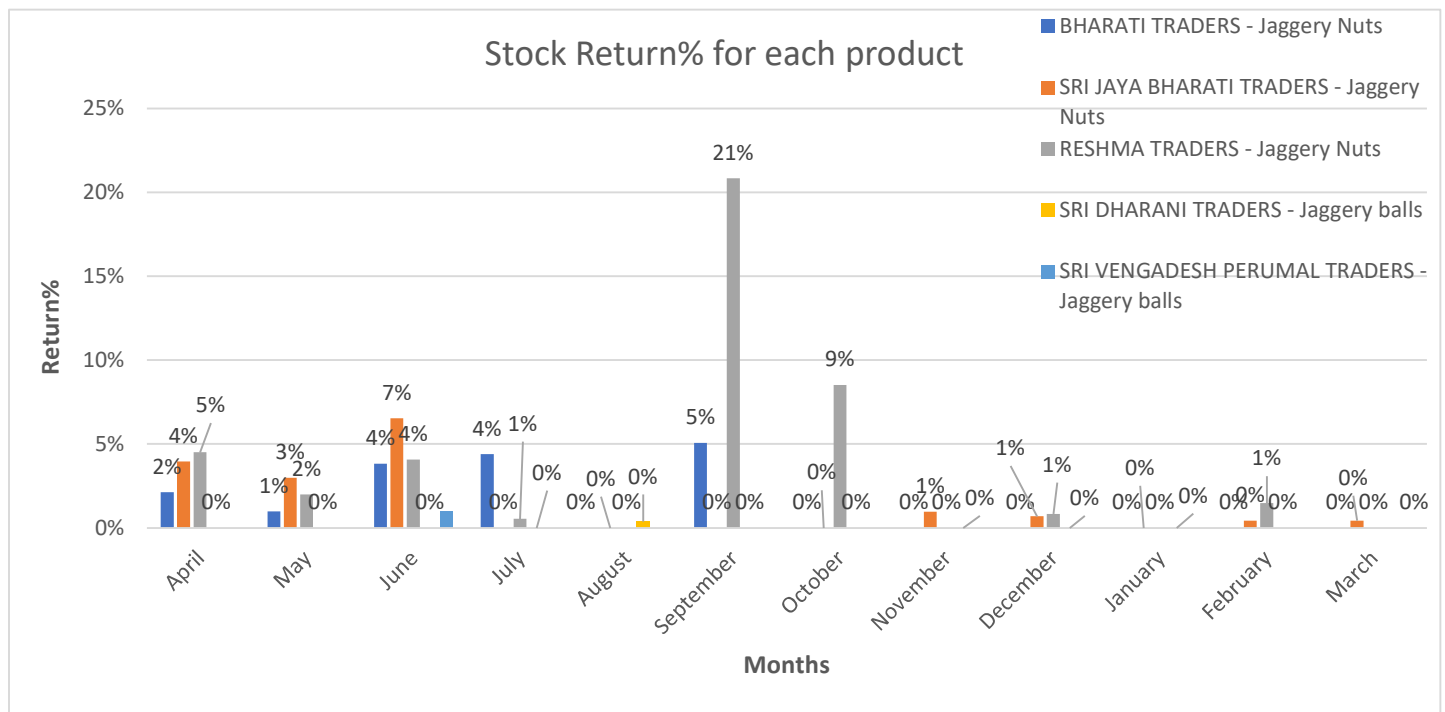


Fig 3.14

From fig 3.14 it is quite evident Reshma Traders has a higher Stock return % (ratio of stock return to sales). Highly Volatile Returns, returns found in almost all months, with a major spike in October (15) and November (23). Some of the highest monthly return rates (21% in October, 9% in November). This suggests significant product quality or batch consistency issues, especially during those months. Could be due to improper storage, supply disruptions, or unreliable sourcing. Adulteration of inorganic products during these months could also lead to such discrepancies.

4. Interpretation of Results and Recommendations

The analysis reveals clear, actionable patterns in sales, revenue, inventory, and pricing for the company's jaggery products. Major findings indicate:

Sales and Revenue Volatility: Core SKUs such as Bharati Traders – Jaggery Nuts and Sri Jaya Bharati Traders show high monthly variability, with strong peaks during festival and competitive pricing periods but steep declines sometimes to zero when pricing or availability are misaligned. This high volatility results in unpredictable revenue streams and highlights a vulnerability to market cycles and competitor pricing.

Inventory Imbalances: Evidence of both stock-outs (missed sales opportunities) and overstocking (tied-up capital and risk of spoilage) reflect shortcomings in forecasting and inventory management, with some months of zero availability and others of excess, low-turnover inventory.

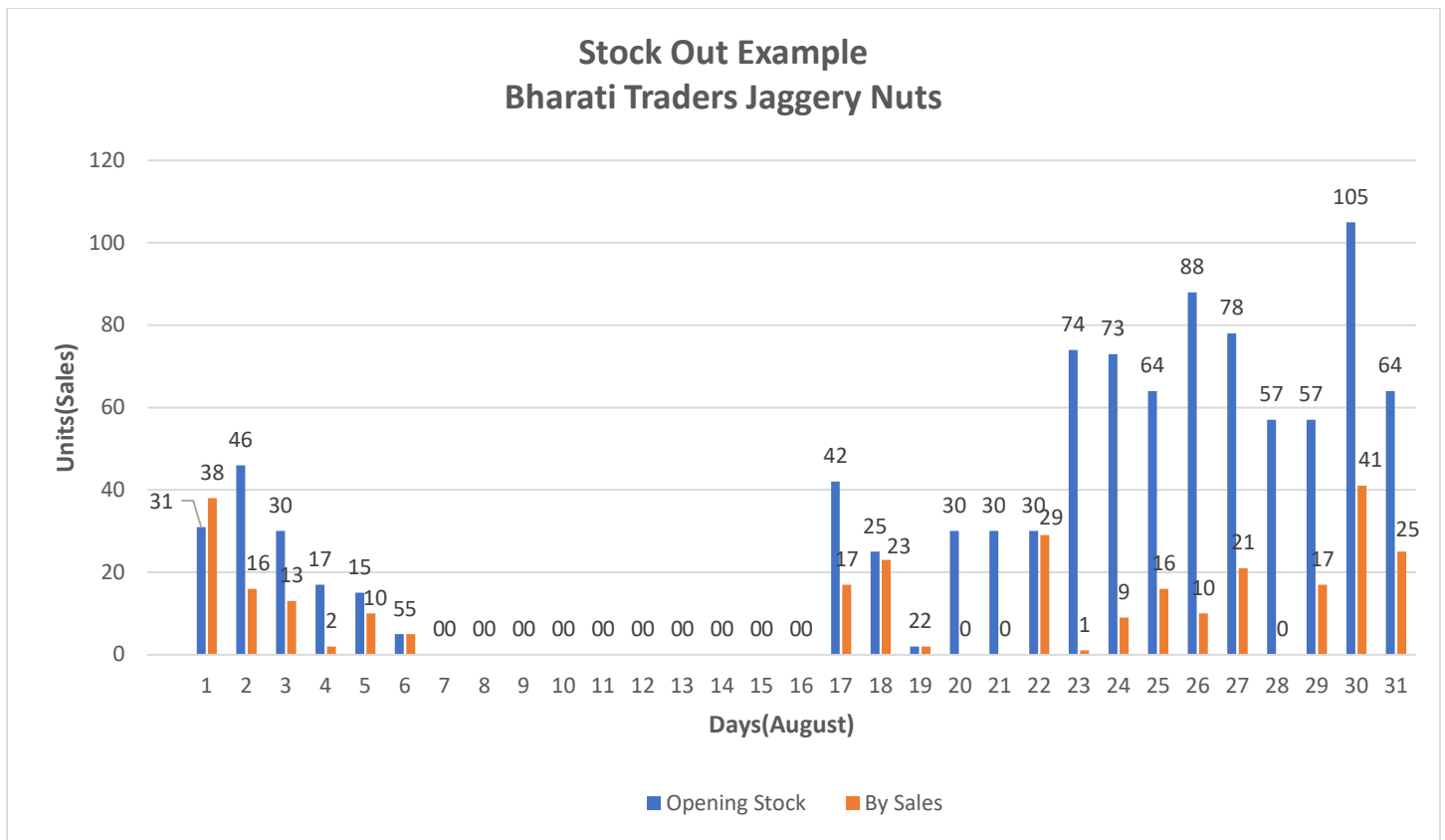


Fig 4.1

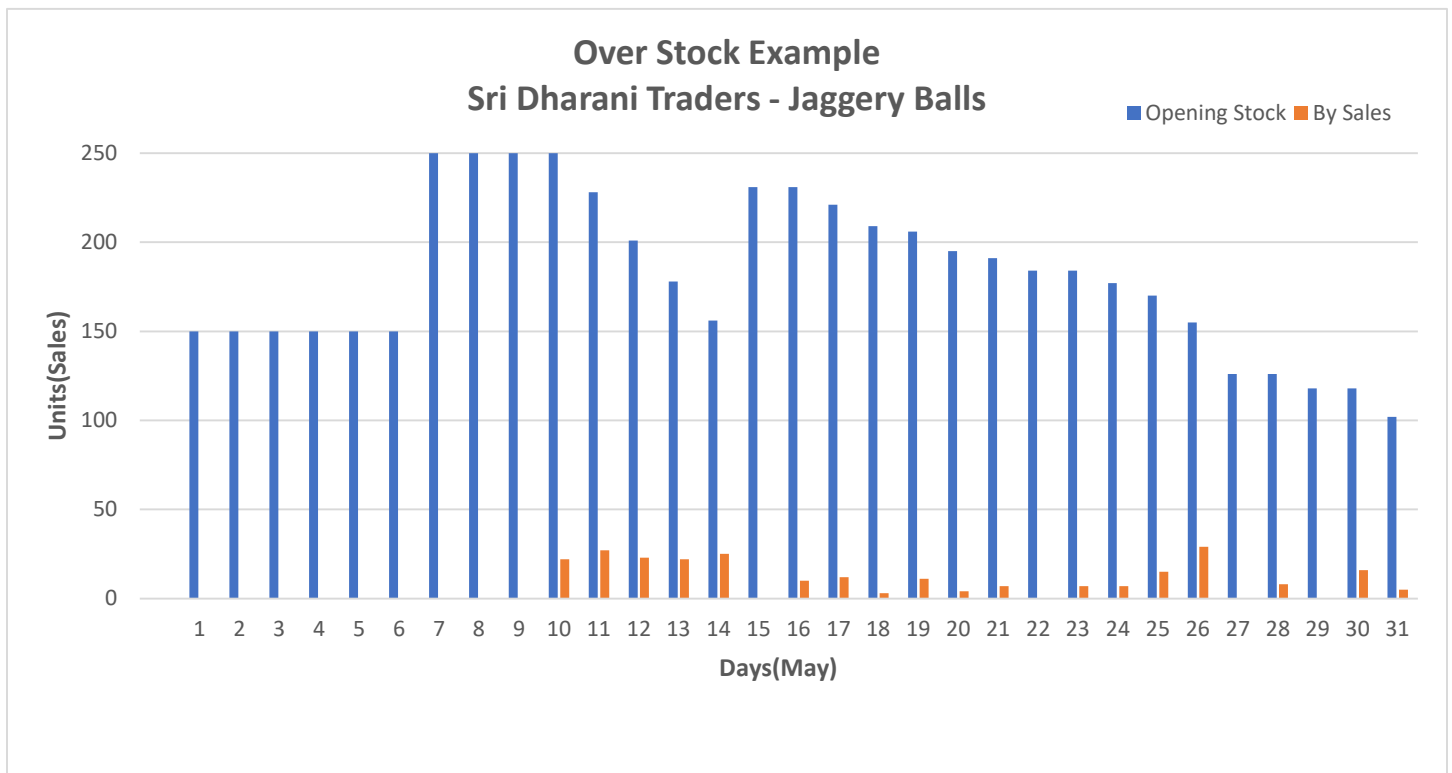


Fig 4.2

From fig 4.1 & 4.2:

During the days from August 1-6 Bharati traders(Jaggery Nuts) ensures daily seasonal sales trend. August 7 - 16 shows a sudden disappearance (due to no opening stocks). This is an example of Stock-out.

From May 1-10, the sales of Sri Dharani traders (Jaggery Balls) have been zero though there is much larger inventory of that brand. This is an example of Over Stock.

This leads to tied-up storage space and capital, hence resulting in revenue instability.

Price Sensitivity and Competitor Alignment: Sales for all major SKUs(especially Bharati and SJB traders) are closely correlated with competitive pricing. When company prices match or slightly undercut the market, sales improve; when prices exceed competitors, sales drop or vanish. Months of non-competitive pricing or absence from the market correspond directly with zero or low sales.

Product Quality and Returns: SKUs with relatively high or volatile return rates (e.g., Reshma Traders) suffer from product quality or batch consistency issues, directly impacting customer satisfaction and repeat business, while others achieve near-zero returns and more stable sales.

ABC Product Classification: A small set of SKUs ("A" category) consistently delivers the majority of sales and revenue demonstrating the importance of focused management on key product lines for revenue stabilization.

Actionable Recommendations

Problem 1: Revenue Instability and Sales Volatility

Implement Real-Time Pricing Intelligence:

Use competitor price monitoring tools to dynamically adjust prices, especially for “A” products, so prices are always at or below market within a 1–2% band. Deploy and test in peak season (within 3 months); aim to reduce sales volatility by 30% in six months.

Problem 2: Inventory Imbalance – Stock-Outs and Overstock

Adopt Data-Driven Demand Forecasting:

Use past sales data, seasonality, and event mapping to set monthly or weekly inventory targets for each SKU. Automate replenishment reminders and set minimum stock thresholds for “A” SKUs. Achieve less than 10% stock-out days and under 15% overstock days for A and B products within the next sales year.

Problem 3: Product Quality and Returns

Strengthen Supplier Quality;

Introduce batch-wise quality checks, especially for high product return SKUs (“C” category). Aim to cut product return rates by at least 50% in the next 12 months for SKUs with monthly returns.

Problem 4: Focus on High-Impact SKUs (ABC Analysis)

Consolidate and Prioritize A and B Products:

Focus marketing, inventory, and pricing efforts on “A” and “B” SKUs, reducing procurement frequency and

working capital tied up in “C” SKUs. Maintain 90% stock availability of “A” products and reduce procurement of “C” products by 25% over one year.

Urgent (Within 3 months): Implement real-time pricing intelligence and inventory monitoring for A-category products.

Long-Term (6–12 months): Establish a cycle of demand forecasting, detailed supplier quality audits, and a quarterly review of ABC product priorities.

Implementing these recommendations is expected to smooth revenue streams, reduce lost sales and excess inventory, lower product return rates, and improve customer retention. With better inventory management and precise pricing strategies, the business will not only stabilize but also be well-positioned for strategic growth.

The Link to [Google Colab](#), analysis done using python.